

MATH 3312, SPRING 2026: SOLUTIONS TO HOMEWORK 3

(1) Determine if the following statements are true or false. Give a line or two in justification.

- (a) There exist distinct irreducible polynomials $f(x), g(x), h(x) \in \mathbb{F}_p[x]$ with leading coefficient 1 such that

$$f(x)g(x) = h(x)^2.$$

Solution. False. This would contradict uniqueness of factorization into irreducibles; see Lecture 7.

- (b) A non-zero polynomial $f(x) \in k[x]$ (k a field) is irreducible if and only if the quotient ring $k[x]/(f(x))$ is an integral domain.

Solution. True. If $f(x)$ is *reducible*, then $k[x]/(f(x))$ cannot be an integral domain: A factorization $f(x) = g(x)h(x)$ into lower degree polynomials will show that $g(x) + (f(x))$ is a non-trivial zero divisor in $k[x]/(f(x))$. Conversely, if $f(x)$ is irreducible, then $k[x]/(f(x))$ is a field and hence an integral domain.

- (c) If $f(x) \in \mathbb{Z}[x]$ is irreducible, then $\mathbb{Z}[x]/(f(x))$ is a field.

Solution. False. The polynomial $x - 1 \in \mathbb{Z}[x]$ is irreducible, but as we saw in Lecture 3, the quotient $\mathbb{Z}[x]/(x - 1)$ is isomorphic to $\mathbb{Z}$, which is not a field.

- (d) A finite field k with 13^2 elements contains an element of (multiplicative) order 5.

Solution. False. By Problem 4 on Homework 2, this would only be possible if $13^2 \equiv 1 \pmod{5}$, but we in fact have $13^2 \equiv 3^2 \equiv -1 \pmod{5}$.

- (e) The quotient ring $\mathbb{F}_{19}[x]/(x^2 - x + 1)$ is a field.

Solution. False. This would only be the case if $x^2 - x + 1 \in \mathbb{F}_{19}[x]$ is *irreducible*. By Problem 7 on Homework 2, this would only be the case if $19 \equiv 2 \pmod{3}$, which it is clearly not.

(2) Suppose that $k \leq L$ is a field extension and that $\alpha \in L$. Show that exactly one of the following is true:

- (a) The evaluation homomorphism $\text{ev}_\alpha : k[x] \rightarrow L$ is injective;
 (b) There exists a unique irreducible polynomial $m_\alpha(x) \in k[x]$ with leading coefficient 1 and an isomorphism of field extensions of k

$$k[x]/(m_\alpha(x)) \xrightarrow{\cong} \text{im } \text{ev}_\alpha \leq L.$$

*In case (2), we will say that α is **algebraic over k** . The polynomial $m_\alpha(x)$ is called the **minimal polynomial of α (with respect to k)**. If we want to emphasize the role of k , we will also denote it by $m_{\alpha,k}(x)$.*

Solution. By Fact 2 in Lecture 6, $\ker \text{ev}_\alpha \leq k[x]$ is an ideal.

By the Theorem from Lecture 7, we have $\ker \text{ev}_\alpha = (f(x))$ for some polynomial $f(x) \in k[x]$.

Now, either $f(x) = 0$, in which case ev_α is injective, or $f(x) \neq 0$. In either case, we have an isomorphism of rings

$$k[x]/\ker \text{ev}_\alpha \xrightarrow{\cong} \text{im } \text{ev}_\alpha \leq L.$$

Since L is a field, we see that the image cannot contain any non-trivial zero divisors (why?). Therefore, $k[x]/\ker \text{ev}_\alpha$ is always an integral domain.

If $f(x) \neq 0$, then 1(b) tells us that $f(x)$ must be irreducible. Therefore, if $\ker \text{ev}_\alpha$ is non-zero, then it is generated by an irreducible polynomial. If $f_1(x)$ and $f_2(x)$ are two polynomials, both generating $\ker \text{ev}_\alpha$, then $f_1(x)$ is a multiple of $f_2(x)$, and *vice versa*. The only way for this to be possible is if $f_2(x)$ is a non-zero constant times $f_1(x)$. Therefore, there is a *unique* irreducible polynomial $m_\alpha(x) \in \ker \text{ev}_\alpha$ with leading coefficient 1 and this generates it.

Note: We can't have any other irreducible polynomials with leading coefficient 1 in $\ker \text{ev}_\alpha$ because any such polynomial would be a multiple of $m_\alpha(x)$.

- (3) Show that the element $\alpha = \sqrt[3]{2} + 2 \in \mathbb{R}$ is algebraic over $\mathbb{Q}$, and compute both $m_{\alpha, \mathbb{Q}}(x)$ and $m_{\alpha, \mathbb{R}}(x)$.

Solution. Clearly, $(\alpha - 2)^3 = 2$. Therefore, α is a zero for the polynomial $(x - 2)^3 - 2 \in \mathbb{Q}[x]$. In other words, $(x - 2)^3 - 2 \in \ker \text{ev}_\alpha$, and so this kernel is non-zero. Thus, α is algebraic over $\mathbb{Q}$.

I claim that $m_{\alpha, \mathbb{Q}}(x) = (x - 2)^3 - 2$. For this, it is enough to know that $(x - 2)^3 - 2$ is irreducible in $\mathbb{Q}[x]$. Since it is a cubic, this is the same as saying that it has no zeros in $\mathbb{Q}$. However, if $\beta \in \mathbb{Q}$ is a zero, then we would have $(\beta - 2)^3 = 2$. This would mean that 2 has a cube root in $\mathbb{Q}$, and this is impossible.

Quick proof: If $\frac{r^3}{s^3} = 2$ with $\frac{r}{s}$ in least terms, then $r^3 = 2s^3$. This implies that r is even. Since $\frac{r}{s}$ is in least terms, s must be odd. But then $2s^3$ is only divisible by 2 while r^3 is divisible by 8.

- (4) Suppose that $k_1 \leq k_2$ is a field extension with k_1 and k_2 both finite.

- (a) Show that every element $\alpha \in k_2$ is algebraic over k_1 .

Solution. There are a couple of ways to see this:

- First, note that, if $|k_2| = p^{d_2}$, then every $\alpha \in k_2$ is a zero for the polynomial $x^{p^{d_2}} - x \in k_1[x]$ (Problem 5 on Homework 2), and so we have $x^{p^{d_2}} - x \in \ker \text{ev}_\alpha$.
- Second, the ring $k_1[x]$ is infinite, while k_2 is finite. Therefore, any homomorphism $k_1[x] \rightarrow k_2$ must have non-zero kernel.

- (b) If $|k_1| = p^{d_1}$ and $|k_2| = p^{d_2}$, show that $d_1 \mid d_2$.

Solution. Choose a generator $\alpha \in k_2^\times$. Then $\text{ev}_\alpha : k_1[x] \rightarrow k_2$ is surjective, and so we have

$$k_1[x]/(m_\alpha(x)) \xrightarrow{\cong} k_2.$$

The left hand side has $(p^{d_1})^m = p^{md_1}$ elements where $m = \deg m_\alpha(x)$. This shows that $md_1 = d_2$, and so $d_1 \mid d_2$.

- (5) Let $f(x) \in \mathbb{F}_p[x]$ be an irreducible polynomial of degree r . Show that the following are equivalent:

- (a) $r \mid d$.
 (b) $f(x)$ is an irreducible factor of $x^{p^d} - x$ in $\mathbb{F}_p[x]$.

Solution. Suppose that $r \mid d$. Then $k = \mathbb{F}_p[x]/(f(x))$ is a field of order p^r . In particular, by Problem 5 on Homework 2, the element $\alpha = x + (f(x)) \in k$ satisfies $\alpha^{p^r} - \alpha = 0$; or, equivalently, $\alpha^{p^r-1} - 1 = 0$. Now, since $r \mid d$, we have $d = rm$ and so $p^d - 1 = p^{rm} - 1$ is a multiple of $p^r - 1$. This follows by plugging $x = p^r$ into the equality

$$x^m - 1 = (x - 1)(x^{m-1} + \cdots + x + 1) \in \mathbb{Z}[x].$$

This in turn implies that $\alpha^{p^d-1} = 1$, and so α is a zero for $x^{p^d-1} - 1$ and hence for $x^{p^d} - x$. All said and done, we find that

$$x^{p^d} - x \in \ker \text{ev}_\alpha = (f(x)).$$

(Why does the last equality hold?)

In other words, $f(x) \mid x^{p^d} - x$. The converse implication will be completed in Homework 4.

- (6) Fix $d \geq 1$. Let $\text{Irr}_{|d}(\mathbb{F}_p)$ be the set of irreducible polynomials $f(x) \in \mathbb{F}_p$ with leading coefficient 1 such that $\deg f(x)$ divides d . Show that we have

$$x^{p^d} - x = \prod_{f(x) \in \text{Irr}_{|d}(\mathbb{F}_p)} f(x) \in \mathbb{F}_p[x]$$

Hint: The key is to know that there are no repeated factors. How would you check if a polynomial has repeated irreducible factors?

Solution. By unique factorization of polynomials (see Lecture 7), we know that we have

$$x^{p^d} - x = \prod_i f_i(x) \in \mathbb{F}_p[x]$$

where $f_i(x) \in \mathbb{F}_p[x]$ are uniquely determined (up to permutation) irreducible polynomials with leading coefficient 1. By the previous problem, the set of polynomials appearing in this factorization are *exactly* the polynomials in the set $\text{Irr}_{|d|}(\mathbb{F}_p)$.

Set $h(x) = x^{p^d} - x$. As the hint says, what we now need to establish is that none of them appears more than once in the factorization. By the criterion from Lecture 9, this can happen precisely when $\gcd(h(x), h'(x)) = 1$.

But we have $h'(x) = p^d x^{p^d-1} - 1 = -1 \in \mathbb{F}_p[x]!$

- (7) Use the previous problem to show that there exist irreducible polynomials in $\mathbb{F}_p[x]$ of every possible degree and hence establish the existence of finite fields of all possible prime power orders.

Solution. For $r \geq 1$, let $\text{Irr}_r(\mathbb{F}_p)$ be the set of irreducible polynomials in $\mathbb{F}_p[x]$ of degree r with leading coefficient 1, and let $N_r = |\text{Irr}_r(\mathbb{F}_p)|$ be its size. Then computing degrees from the equality in Problem 6 tells us:

$$p^d = \deg(x^{p^d} - x) = \sum_{f(x) \in \text{Irr}_{|d|}(\mathbb{F}_p)} \deg f(x) = \sum_{r|d} \sum_{f(x) \in \text{Irr}_r(\mathbb{F}_p)} r = \sum_{r|d} r N_r.$$

This tells us that

$$d N_d = p^d - \sum_{r|d, r \neq d} r N_r$$

So to show that $N_d \neq 0$, it is enough to know that $p^d > \sum_{r|d, r \neq d} r N_r$. But note that the same equation with d replaced by r tells us that $r N_r \leq p^r$. Therefore, we see

$$\sum_{r|d, r \neq d} r N_r \leq \sum_{r|d, r \neq d} p^r < \sum_{r=0}^{d-1} p^r = \frac{p^d - 1}{p - 1} < p^d,$$

as desired.

You can actually use Problem 6 to establish a formula for the number of irreducible polynomials of any degree.